



Project Management System

FUNCTIONAL REPORT

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1. Project Overview

The BPCL Project Management System is a centralized, role-based web application designed to track, manage, and monitor projects, milestones, and cross-team collaborations within Bharat Petroleum Corporation Limited. By providing distinct dashboards tailored for Administrators, Managers, Department Heads, and Team Members, the system ensures that every stakeholder has immediate access to relevant data, reducing friction and improving cross-departmental coordination.

The platform is built around secure, backend-enforced authentication and access control, a clean and uncluttered interface, a cross-project activity feed for staying on top of updates, and dashboards that work smoothly across devices.

Key Features at a Glance

- Real-time project progress tracking with visual completion indicators
- Granular milestone assignments tied to specific teams and deadlines
- Automatic “Delayed” flagging for overdue pending milestones
- Rich communication logs with manager-flagging capabilities
- Role-specific dashboards: Admin, Manager, Department Head, and Team Member
- Session security with JWT-based tokens and automatic re-login on expiry
- A global Recent Updates feed showing the latest activity across every project and location
- Dynamic Location & Department filters available to Managers as well as System Admins
- A simplified, decluttered interface — secondary actions consolidated under a single overflow (...) menu
- Clean, well-formatted Excel export for offline reporting and stakeholder meetings
- An interactive, airgap-safe API documentation hub (Swagger UI) for developers
- Advanced global search across projects, milestones, and users
- Full role-based access control (RBAC) enforced at the backend API level, not just in the UI

Target Audience: This document is intended for end-users, project managers, and system administrators at BPCL who will interact with the Project Management System.

2. Website Structure (Pages)

The application is structured into the following core views to optimize navigation and data visibility across different user roles:

Page	Purpose
Login / Registration	Entry point for all users. New users register and await Admin approval. Existing users are routed to their role-specific dashboard automatically.
Admin Dashboard	High-level administrative interface for creating, editing, and deleting system entities: Users, Teams, Projects, and system dropdown options. Distinguishes System Admin (full access) from Team Admin (scoped to their own team).
Manager Dashboard	Overview of projects assigned to the manager’s team(s). Prioritizes project health (Delayed vs. Pending) and team communications.

Page	Purpose
Team Member Dashboard	Focused personal workspace where users only see milestones assigned to their team, keeping execution clear and distraction-free.
Project Detail Page	Deep-dive interface for a single project. Features a scrollable left panel for project info and links, and a milestone+log panel for tracking and communication.
Recent Updates	Global, read-only feed grouped by Location showing the latest logged update for every project the user can see. Replaces manual project-hopping when catching up on activity.
Team Overview	Head/Admin-only dashboard grouping every team by Location (refinery), showing project counts, completed/pending/delayed milestone counts, and progress at a glance.
Search Hub	Globally accessible advanced search page that dynamically queries projects, milestones, and users for instant data retrieval.
Profile	Personal account page for viewing organization details, assigned teams, roles, and changing your password.

Navigation Architecture

A persistent Top Navigation Bar is accessible from every page, providing:

- Dashboard / Recent Updates / Search / Profile — primary navigation tabs
- Location filter — filter by region (e.g., MR, BR, KR)
- Department filter — filter by business unit (e.g., Digital, APC, IS)
- Project Type filter — filter by classification (e.g., BP, NPCB)
- Log Filter (Managers/Heads only) — toggle to view only flagged logs
- Search access — global Search Hub from any page
- Profile menu — user identity, active role badges (e.g., Manager / Admin), and sign-out

Interface simplification: project cards and detail pages keep every capability available while grouping secondary actions (Edit, Export, Chart, Close, View, Eye, Delete) under a single “...” overflow menu or compact icon row, keeping the interface clean and easy to scan.

3. Role-Based Access & Functional Capabilities

The system strictly segregates responsibilities across four primary roles, plus a Team Admin sub-scope for Managers. Each role’s permissions are enforced both in the UI and independently re-validated on every backend API call.

Role	Key Capabilities
Admin (System)	Create, edit, delete, and assign roles to system users • Create, edit, and delete functional teams • Define parent projects, assign project managers, and create top-level milestones • Modify system dropdown options: Locations, Departments, Project Types • Access the full Team Overview across all locations
Team Admin (Manager)	Same CRUD tools as System Admin, scoped only to their own assigned team(s) and projects • Cannot see or modify teams/projects outside their scope
Manager	View projects assigned to their own team(s) • Monitor milestone statuses — pending

Role	Key Capabilities
	milestones past their deadline auto-display a red Delayed badge • Read all communication logs and flag critical entries for immediate attention • Use interactive Location and Department dropdown filters on the Team Overview page • Export comprehensive Excel reports per project or across all teams
Department Head	View all projects across all teams, locations, and departments • Monitor milestone statuses across all departments • Read all communication logs and flag critical entries • Use the Recent Updates and Team Overview pages for a cross-organization view • Export comprehensive Excel reports across all teams and departments
Team Member	View only the milestones assigned to their specific Team • Update progress status of assigned milestones (Pending to Completed) • Add daily progress logs, notes, or blockers to milestones • Edit and delete their own log entries

4. Step-by-Step User Guide

4.1 Authentication

Logging In

1. Navigate to the BPCL Project Management System homepage.
2. Enter your BPCL username (or email) and password in the Sign In form.
3. Click the Sign In button. The system authenticates against the backend and issues a secure session token; you are automatically routed to your role-specific dashboard.

Session Security: every session is governed by a JSON Web Token (JWT) with a 24-hour expiration window. If a dashboard is left open and the token expires, the next action automatically triggers a 401 response, and the app securely redirects you back to the Login screen to re-authenticate — preventing stale sessions from mutating data.

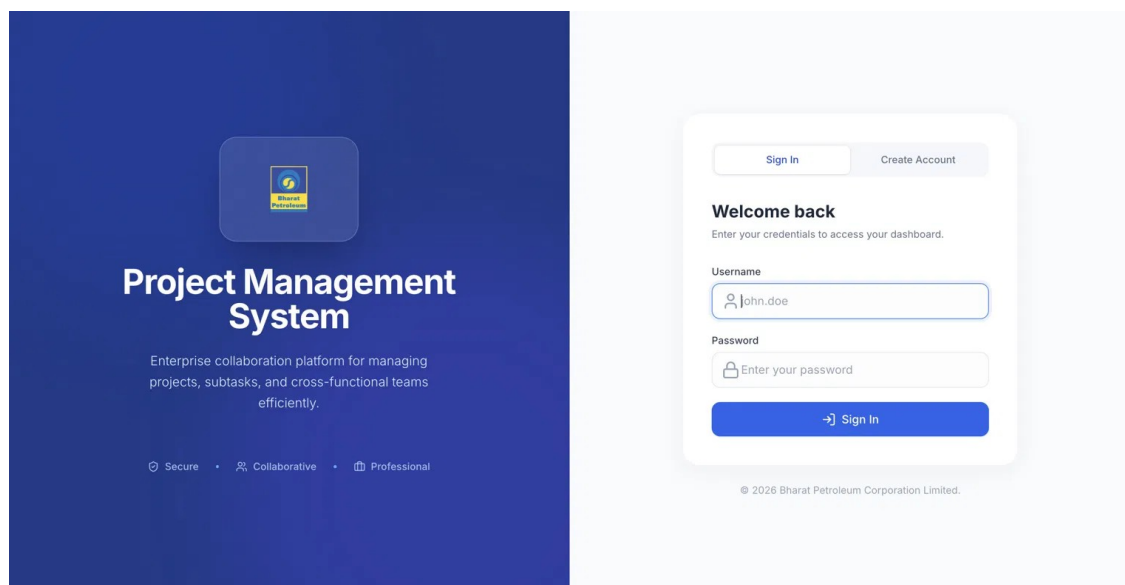


Figure 1 — Login / Sign In Screen

Creating an Account

4. Click the Create Account tab on the login page.
5. Fill in your full name, username, password, and optionally select a Team, Location, and Department.
6. Submit the form. Your account is placed in a Pending state until an Administrator approves and assigns your role.

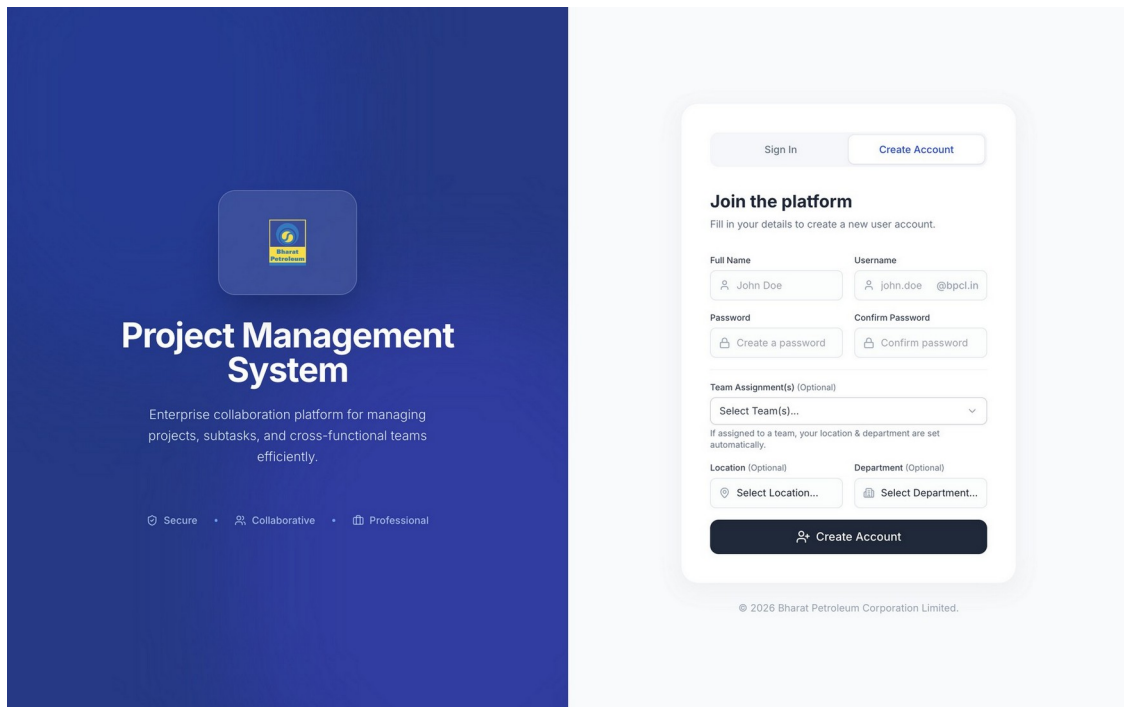


Figure 2 — Create Account Screen

4.2 Admin Operations

The Admin Panel clearly separates System Admin capabilities from Team Admin (Manager-scoped) capabilities, shown via a badge in the panel header.

Team Admin View

A Manager who also holds Team Admin rights sees a scoped panel limited to their own team(s) and projects: creating projects, assigning managers, and setting Location/Department/Project Type for their team only.

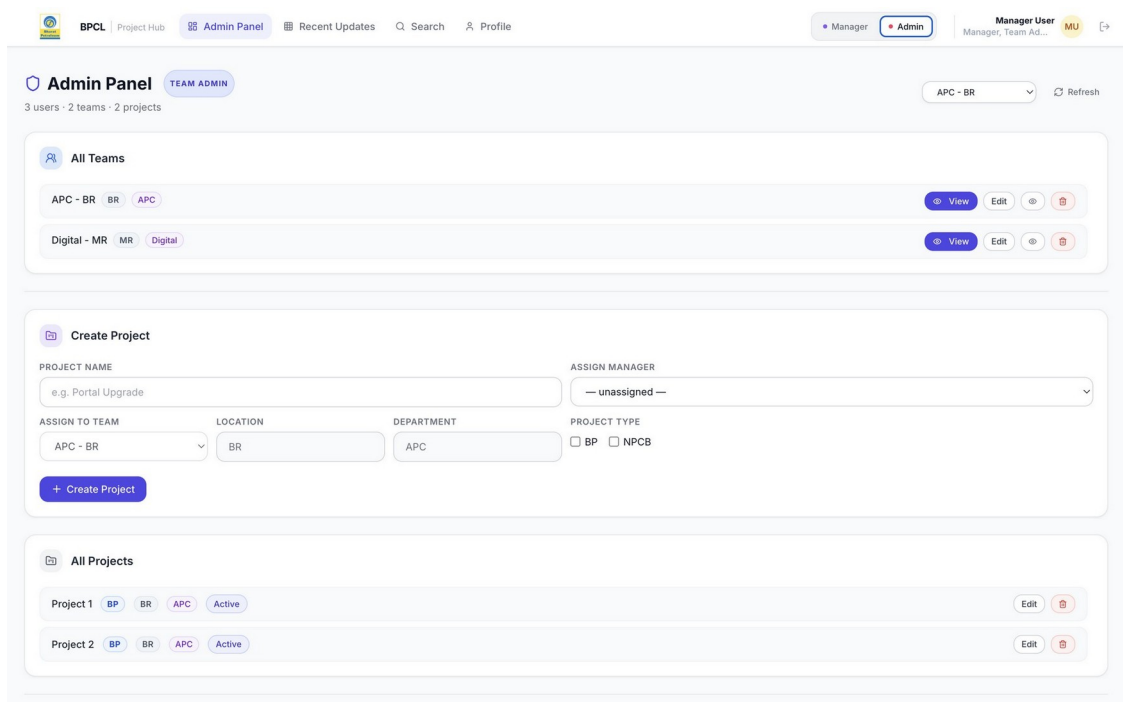


Figure 3 — Admin Panel, Team Admin scope

System Admin View

7. Navigate to the Admin Panel (System Admin badge shown at top).
8. Under Create Team, enter Team Name, Location, and Department, then click Create Team.
9. Under All Teams, use the compact View / Edit / Eye (visibility) / Delete icon row to manage any team system-wide.
10. Under Create Project, enter the Project Name, assign a Manager, and set Location, Department, and Project Type, then click Create Project. The project becomes immediately visible to its Manager.

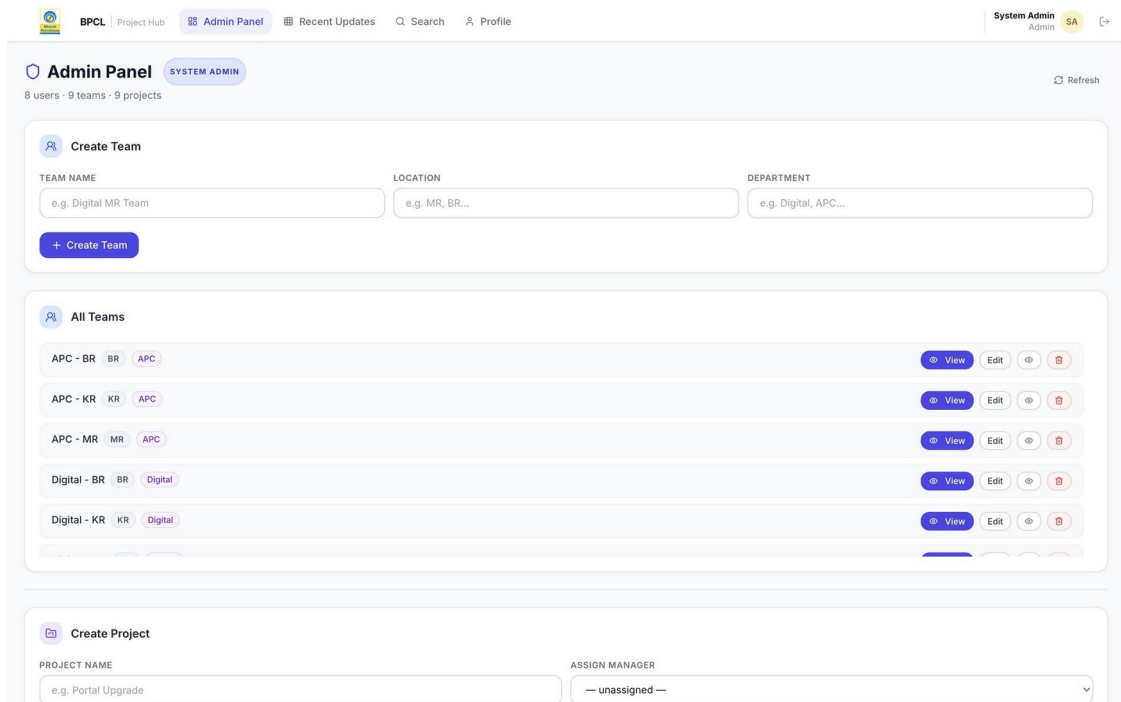


Figure 4 — Admin Panel, System Admin scope

Adding Milestones (Subtasks)

11. Open a project's detail page and use the Create Project Milestone panel.
12. Enter the Milestone Name and assign it to a specific Team.
13. Set the Start Date and Deadline.
14. Click Create. The milestone now appears on the assigned team's dashboard.

Managing Users & Teams

- Users: Edit roles, activate pending accounts, or remove users from the system.
- Teams: Create functional teams tied to locations and departments; edit or delete as needed.
- System Options: Add or remove entries from Location, Department, and Project Type dropdown lists.

API Documentation Hub

A fully interactive Swagger UI is available to developers at <http://<server-address>/api/docs/>, mapping all 21 system endpoints under the OpenAPI 3.0.3 standard. Developers can log in via `POST /api/login`, copy the returned token, click Authorize, paste “Bearer <token>”, and execute live

API calls directly from the browser — no Postman required. The hub works fully offline, which is required for BPCL’s airgapped server environments.

4.3 Manager Dashboard & Oversight

After logging in as a Manager, you land on the Manager Dashboard showing all projects assigned to your team(s).

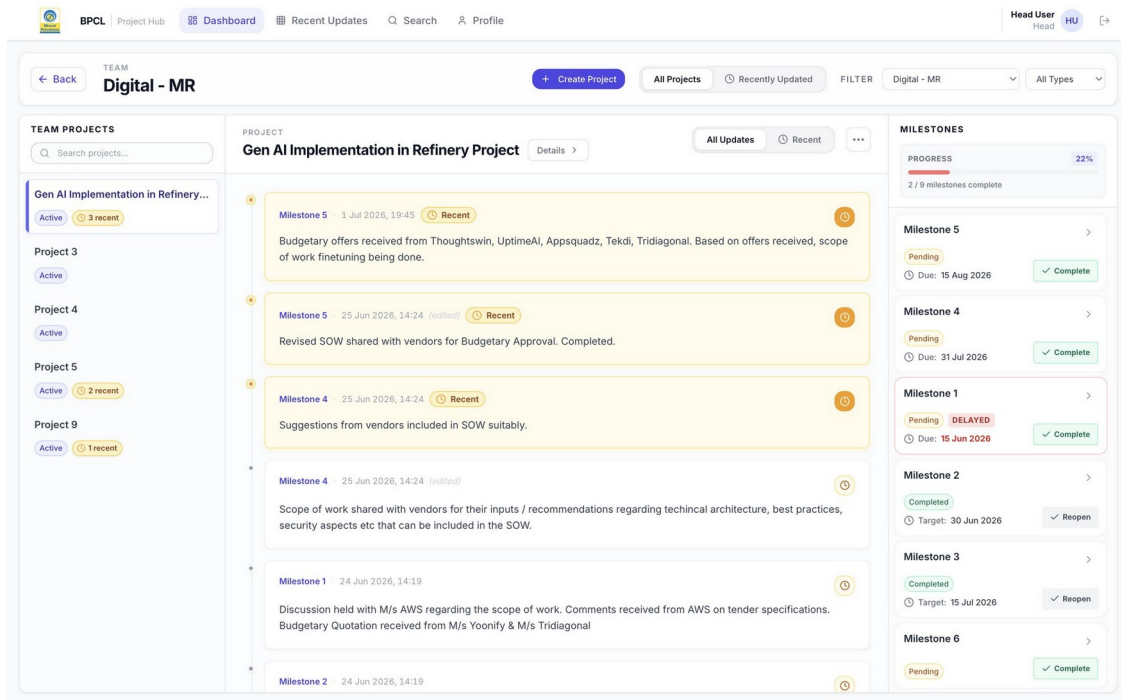


Figure 5 — Manager Dashboard with project detail and milestone panel

Understanding the Dashboard

- Each Project Card in the left rail displays the project name, a status pill (e.g., Active), and a condensed “X recent” activity badge.
- A red Delayed badge appears on any milestone whose deadline has passed while still Pending.
- The Recent badge highlights logs added within the last few days for quick attention.
- The Milestones panel on the right shows per-milestone progress with one-click Complete / Reopen controls.

Filtering Projects

- Use the Location, Department, and Project Type dropdowns in the top bar to narrow your view.
- Use the All Updates / Recent toggle to show only recent log activity.

Exporting Reports

- Single Project Export: use the overflow (...) menu on a project card to download an .xlsx file with all milestones, teams, deadlines, and the full communication log history.
- Team Overview Export: use Export on the Team Overview / Recent Updates page to generate a multi-sheet, well-formatted Excel file — one sheet per team — covering all assigned projects, pending milestones, and performance metrics. Column widths, headers, and status coloring are designed for easy reading in offline reports.

4.4 Project Detail Page (All Roles)

Clicking on a project opens the Project Detail Page, a two-panel view designed for deep inspection and communication.

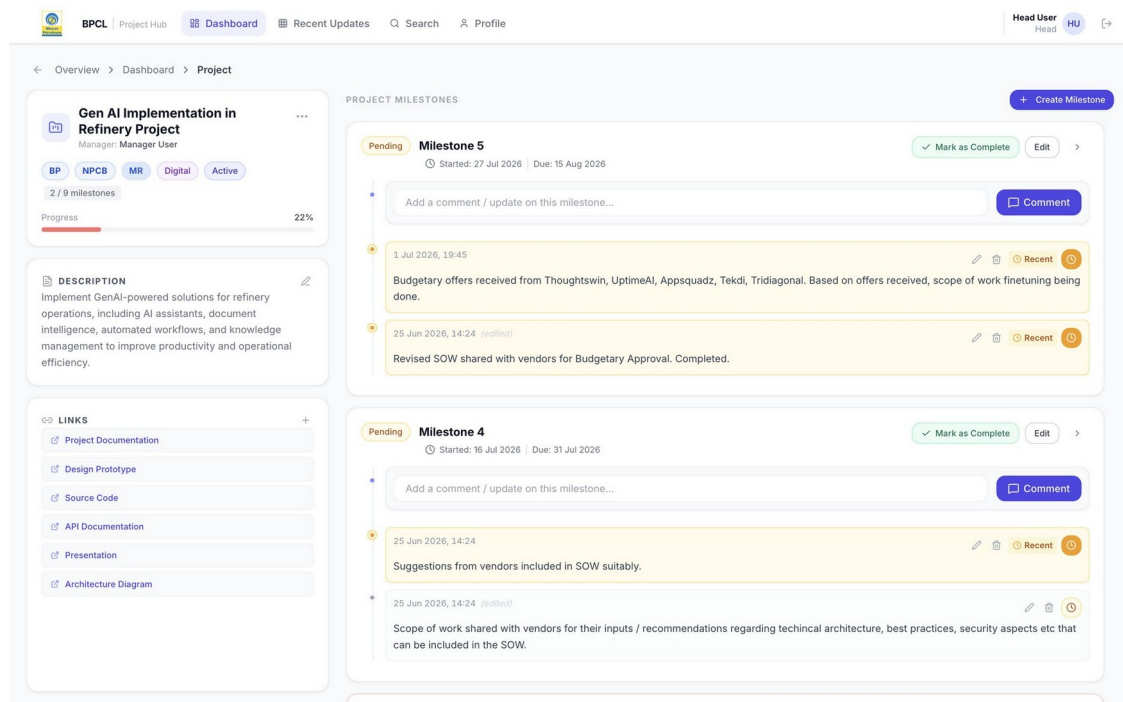


Figure 6 — Project Detail Page with milestone list and activity logs

Left Panel — Project Info

- Displays the project name, manager, type tags, description, and progress percentage.
- Shows all attached links (Project Documentation, Design Prototype, Source Code, API Documentation, Presentation, Architecture Diagram).
- Admin/Manager actions (Edit, Export, Chart, Close) are consolidated behind the “...” menu next to the project title, keeping the header clean.

Right Panel — Milestones & Logs

- Lists all milestones with their status (Pending / Completed / Delayed), start date, and deadline.
- Each milestone has an expandable activity log thread below it, with edit/delete/history icons shown inline.
- Team Members can type in the comment box and click Comment to post an update.
- Managers/Admins see “Mark as Complete” directly on each pending milestone card.

4.5 Milestone Detail Page

Clicking the arrow on any milestone card opens its dedicated Milestone Detail Page.

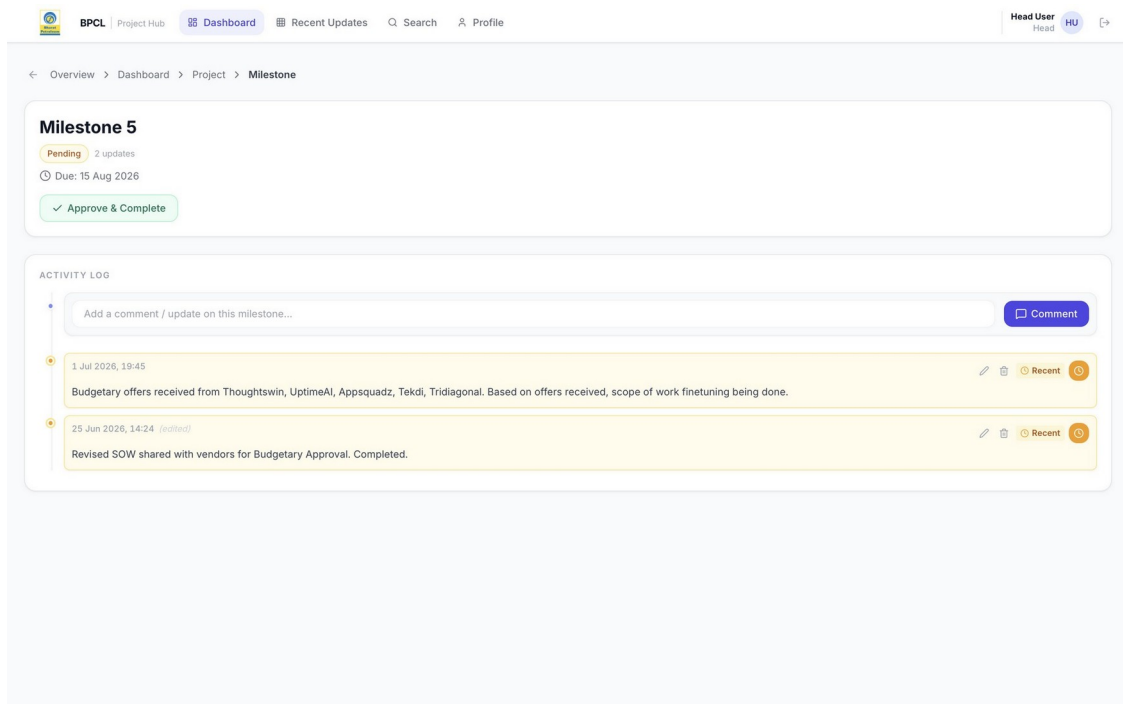


Figure 7 — Milestone Detail Page with activity log

- Displays the milestone name, status badges (Pending, Delayed), due date, and update count.
- The Activity Log shows all comments in chronological order, each with edit/delete/history controls.
- Managers/Admins see the Approve & Complete button to mark a milestone done.
- Team Members can add, edit, or delete their own comments.

4.6 Search Hub

The Search Hub, accessible from the top navigation bar, provides a powerful global search across the entire system, presented as a browsable grid of Active Projects alongside the search box.

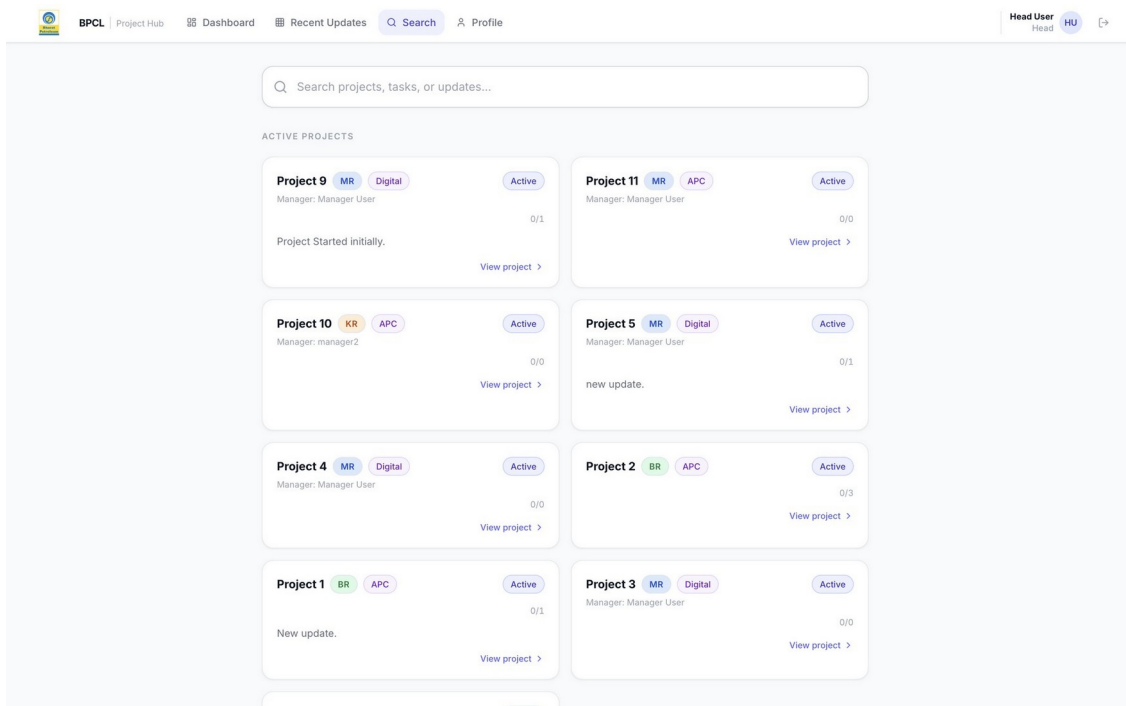


Figure 8 — Search Hub with Active Projects grid

- Search by project name, milestone name, or username — results update dynamically as you type.
- When no query is entered, all Active Projects visible to the user are shown as cards with tags, assigned Manager, completion count, and the latest update snippet.
- Click any result or “View project” to navigate directly to that project, milestone, or user profile.

4.7 Recent Updates Page

The Recent Updates tab in the top navigation gives every role (scoped to what they can see) a single, scrollable feed of the latest activity across all projects, grouped for clarity by Location.

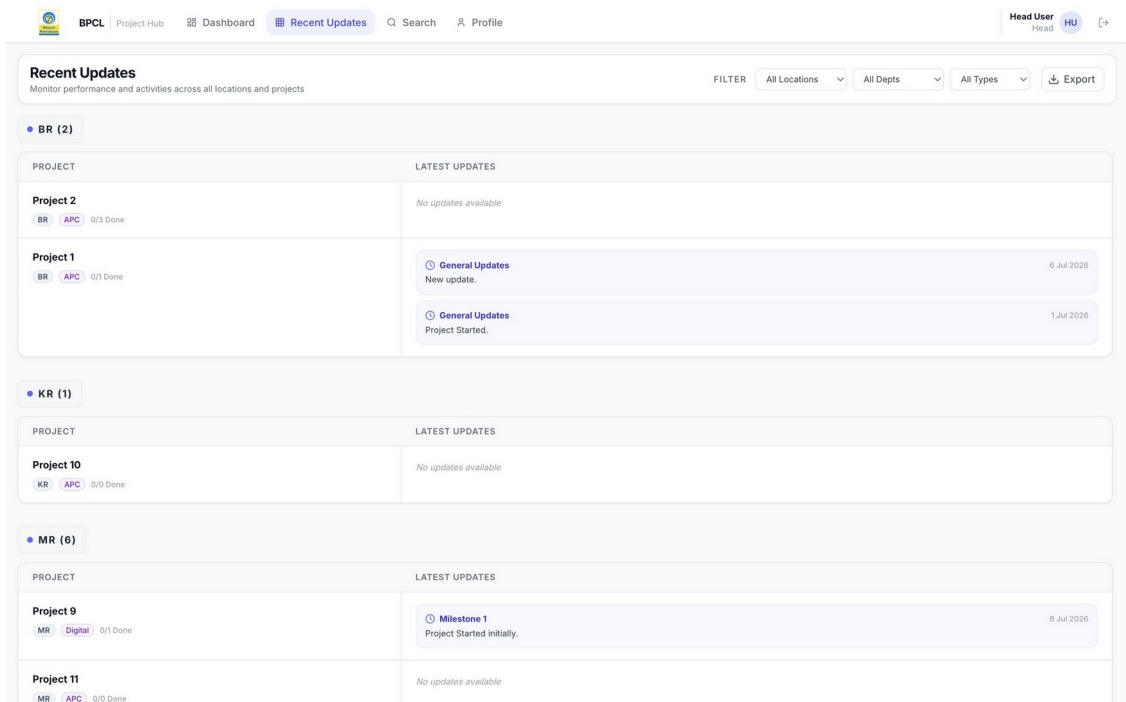


Figure 9 — Recent Updates page, grouped by Location

- Projects are grouped under collapsible Location headers (e.g., BR, KR, MR) with a project count per group.
- Each row shows the Project (name, location/department tags, completed-vs-total milestone count) alongside its Latest Updates — the most recently flagged or most recent log entries, with edited/recent indicators and timestamps.
- Location, Department, and Project Type filters at the top narrow the feed instantly.
- An Export button downloads the filtered view as an Excel report.
- Managers and Team Members automatically see only projects tied to their own team(s); Admins and Department Heads see everything.

4.8 Team Overview Page

Available to Department Heads and System Admins, Team Overview groups every functional team by Location (refinery) so leadership can compare performance across sites at a glance.

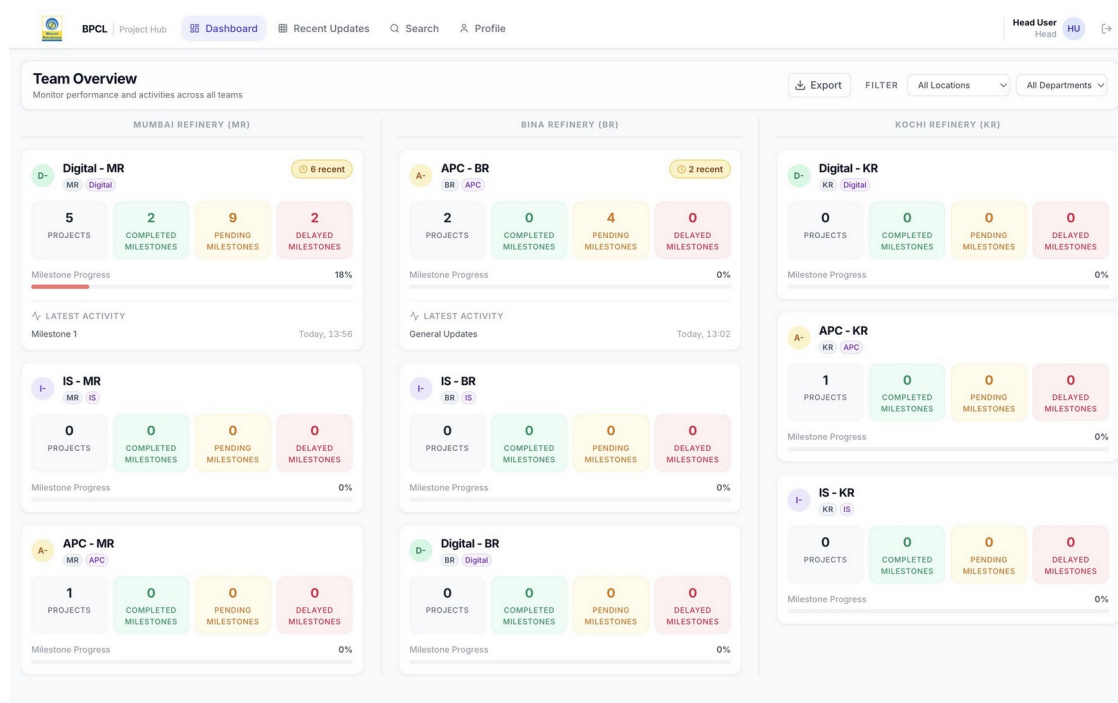


Figure 10 — Team Overview grouped by refinery (Mumbai / Bina / Kochi)

- Teams are grouped into columns by Location (e.g., Mumbai Refinery (MR), Bina Refinery (BR), Kochi Refinery (KR)).
- Each team card shows Projects, Completed Milestones, Pending Milestones, and Delayed Milestones as at-a-glance counters, plus a milestone progress bar and the latest activity timestamp.
- All Locations / All Departments filters at the top let a Head or Admin narrow the view; an Export button produces the multi-sheet Excel overview.

4.9 Team Member Dashboard

Team Members land on a personal Team Dashboard focused on their own team's active projects and milestones.

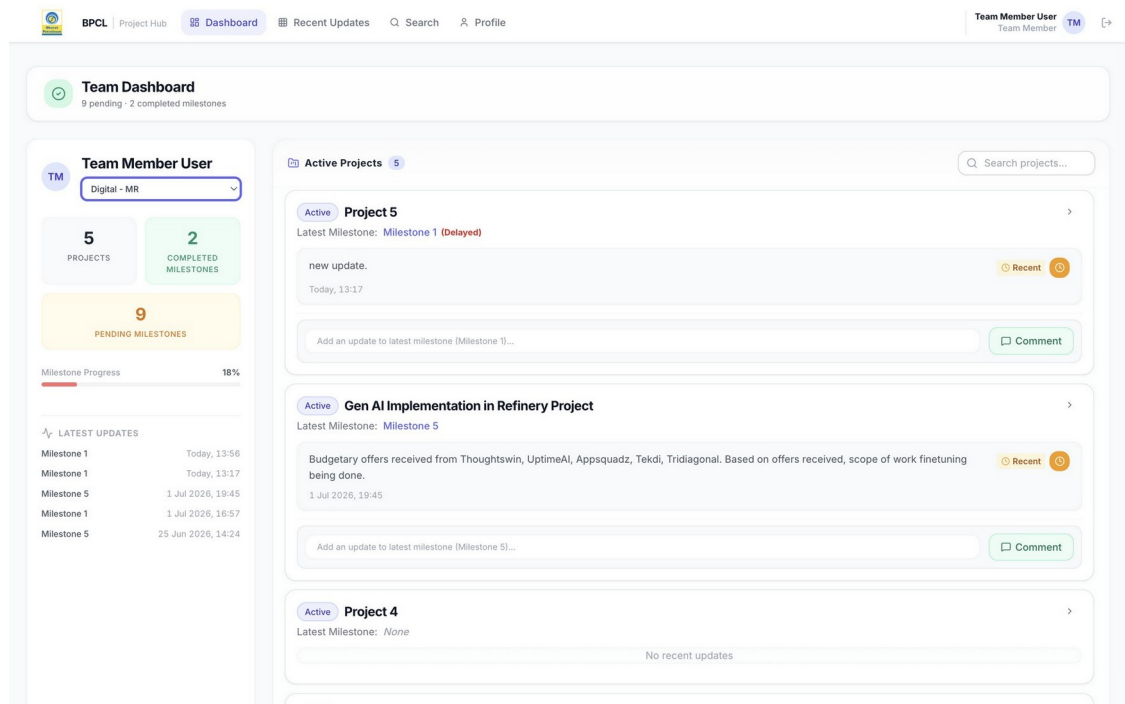


Figure 11 — Team Member Dashboard

- Summary tiles show Projects, Completed Milestones, and Pending Milestones, plus an overall Milestone Progress bar.
- A Latest Updates list on the left surfaces the most recent activity across the user's projects.
- Each Active Project card on the right shows its latest milestone and lets the Team Member post a Comment update directly, without leaving the dashboard.

4.10 Profile Page

Every user can manage their own account details from the Profile page, accessible from the top navigation bar.

The screenshot shows the 'My Profile' page in the BPCL Project Hub. The page has a header with navigation links: BPCL, Project Hub, Dashboard, Recent Updates, Search, and Profile. The user is logged in as 'Head User' (HU). The profile page is titled 'My Profile' and includes a subtitle 'Manage your personal information and security.' The form is divided into several sections:

- PERSONAL DETAILS:** Includes 'Full Name' (Head User) and 'Email Address' (head.user@bpcl.in).
- ORGANIZATION:** Includes 'Location' (All), 'Department' (All), and 'Assigned Team(s)' (Select Team(s)). A note states: 'Team assignment defines which projects your logs contribute to.'
- ROLES:** Includes a dropdown menu showing 'Head'.
- CHANGE PASSWORD:** Includes fields for 'New Password' and 'Confirm New Password'. A note states: 'Leave these fields blank if you do not want to change your password.'

A 'Save Changes' button is located at the bottom right of the form.

Figure 12 — My Profile page

- Personal Details: Full Name and Email Address.
- Organization: Location, Department, and Assigned Team(s) — team assignment determines which projects a user's logs contribute to.
- Roles: displays the user's assigned role(s), e.g., Head, Manager, Team Admin.
- Change Password: users may update their password at any time; fields are left blank to keep the existing password.

5. Status & Badge Reference

The system uses visual badges and color codes to communicate project and milestone health at a glance.

Badge	Color	Meaning
Pending	Amber	The milestone has not yet been completed. Work may be in progress.
Completed	Green	The milestone has been marked complete by a Team Member and approved.
Delayed	Red	The milestone is still Pending after its deadline has passed.
Active	Blue	The project is currently active and accepting milestone updates.
Recent	Amber	A log entry was posted within the last few days — new activity.
Flagged	Amber border	A Manager or Head has flagged this log entry as requiring immediate attention.
System Admin / Team	Purple pill	Indicates the scope of Admin Panel access currently

Badge	Color	Meaning
Admin		active for the logged-in user.

6. Security & Platform Summary

The following backend and platform capabilities underpin the system. They are not separately visible as UI screens but affect every page described above:

- JWT-based sessions (24-hour expiry) with automatic redirect to Login on token expiry, so dashboards never stay open indefinitely with a stale session.
- Full role-based access control re-validated on every API call at the backend, not only hidden/shown in the UI.
- Rate limiting of 100 requests per minute per IP address, protecting the system from scripted abuse or brute-force attempts.
- An airgap-safe, fully offline Swagger UI (OpenAPI 3.0.3) documenting all 21 endpoints.
- A simplified interface: actions are consolidated into compact icon rows and overflow menus across the Admin Panel, Manager Dashboard, and Project Detail Page, keeping the UI clean while retaining full functionality.
- Well-formatted Excel export across all export types (single project, team overview, recent updates).
- Cross-project views — Recent Updates and Team Overview — give Heads, Admins, and Managers a fast way to catch up on activity without opening every project individually.